

## CAPSTONE PROJECT – TOPIC DESCRIPTION

### A Web-Based System for Real-Time Hotel Market Price Monitoring and Forecasting Using Machine Learning

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#### 1. Topic Orientation and Motivation

Hotel room prices on Online Travel Agencies (OTAs) such as Agoda and Booking.com fluctuate continuously under dynamic pricing driven by demand, seasonality, events, and competitor behavior. Large hotel chains manage this volatility with commercial revenue management and rate intelligence platforms (e.g., RateGain, Lighthouse, Duetto), which cost hundreds to thousands of USD per month. However, Vietnam's accommodation sector - over 38,000 establishments, the majority independent small and medium-sized hotels - largely cannot access such tools: competitor rates are checked manually and room prices are set by intuition. On the other side of the market, individual travelers face the mirror problem of price opacity, with no way to track price history or anticipate price movements.

This project develops an end-to-end web-based market price intelligence system for the Vietnamese hotel market that: (1) continuously collects room prices from Agoda and Booking.com across major Vietnamese cities, (2) applies machine learning to forecast short-term price movements (1–14 days ahead), and (3) serves the forecasts through two decision layers built on a shared prediction engine:

**Primary use case - Hotelier / Business view:** a competitor rate intelligence dashboard where a hotel selects its competitive set (comparable nearby hotels), monitors their real-time rates across both OTAs, views market-level price forecasts, receives alerts when its own price deviates significantly from the market or when market movements are predicted (holidays, events), and detects cross-OTA rate disparities.

**Secondary use case - Consumer view:** price history charts and short-term forecasts for individual hotels, supporting travelers' booking-timing decisions.

Both views are powered by a single trained forecasting model; only the application layer differs. This addresses a clear research and market gap: affordable, ML-based rate intelligence for the under-explored Vietnamese market, accounting for local demand drivers (Tet, summer domestic peak, regional festivals) that existing Western-oriented tools ignore.

#### 2. Dataset and Data Collection (Current Progress)

The dataset is self-collected. A web-based crawling tool has already been implemented and is operational. The tool manages lists of hotels organized by region/city; for each hotel and each selected check-in/check-out date pair, it automatically extracts all publicly listed information. Each record contains:

**Price data:** nightly rate (VND), discount, original price, room type, cancellation policy, breakfast inclusion.

**Hotel attributes:** name, star rating, location (city/district), review score, number of reviews, amenities.

**Temporal metadata:** scrape timestamp, check-in date, and lead time (days between observation and check-in) - the key dimension for modeling how prices evolve as the stay date approaches.

The crawler can run automatically at predefined times or be triggered manually on demand by the user. Each run observes a hotel's prices for multiple future check-in dates, producing dense price time series. Data is enriched with external public sources: the Vietnamese holiday/festival calendar, weather forecasts (Open-Meteo API), and VNAT tourism statistics. Target scale: 200–300 hotels (3–5 stars) in 5 cities (Ho Chi Minh City, Hanoi, Da Nang, Nha Trang, Phu Quoc), accumulating 100,000+ price records over the collection period. Supervised-learning labels are derived automatically from

the time series itself (future prices are observed later by the same crawler), so no manual annotation is required.

### 3. Methodology and Models

**Feature engineering:** Four feature groups are constructed: hotel attributes; temporal features (day of week, holiday/festival flags, lead time, season); price-history features (lag values, rolling mean/std, price velocity); and market context (competitive-set average price, cross-OTA price ratio, weather, tourist arrivals).

**Models:** Four models - two traditional machine learning models and two deep learning models - are trained and systematically compared for price forecasting at 1, 3, 7, and 14-day horizons. No model is designated in advance as the primary candidate; the best-performing model is selected empirically based on experimental results.

| Model         | Family         | Paradigm represented                       |
|---------------|----------------|--------------------------------------------|
| XGBoost       | Traditional ML | Boosting (sequential error correction)     |
| Random Forest | Traditional ML | Bagging (variance-reducing ensemble)       |
| LSTM          | Deep Learning  | Recurrent sequence modeling (gated memory) |
| Transformer   | Deep Learning  | Attention-based sequence modeling          |

**Model selection rationale:** Each model represents a methodologically distinct paradigm, making the comparison meaningful rather than redundant. XGBoost represents gradient boosting, the family that consistently dominates tabular prediction benchmarks and prior price-forecasting studies; Random Forest represents bagging, an independent ensemble principle (variance reduction rather than bias reduction). LSTM is the canonical recurrent architecture for temporal dependencies in price series, well-established in tourism demand forecasting literature, while the Transformer represents the current attention-based direction in time-series forecasting - providing a recurrent-vs-attention contrast. Near-identical variants within a family (LightGBM/CatBoost vs. XGBoost; GRU vs. LSTM) are deliberately excluded: their differences are mainly engineering optimizations, adding training cost without methodological insight. Classical statistical models (ARIMA, exponential smoothing) are excluded because they cannot naturally exploit the rich exogenous features central to this problem (hotel attributes, holidays, weather, competitive-set context).

A strict time-based train/validation/test split (70/15/15, chronological) prevents data leakage; hyperparameters are tuned with Optuna; SHAP analysis provides interpretability (which factors drive market price movements - directly valuable insight for hoteliers). Models are evaluated on both price regression and price-direction classification (increase/decrease). The winning model becomes the shared forecasting engine: hotel-level predictions are served directly to the consumer view and aggregated over competitive sets for the hotelier view.

**System:** Scrapy/Selenium crawlers with scheduled and on-demand execution, PostgreSQL + Redis storage, FastAPI backend, and a ReactJS dashboard (competitor rate monitoring, market forecast panel, pricing position alerts, price history charts).

### 4. Evaluation and Expected Results

Evaluation is fully quantitative, on held-out chronological test data: (1) Forecast accuracy - target  $\text{MAPE} \leq 12\%$  for 7-day forecasts, price-direction accuracy  $\geq 80\%$ , with SHAP-based interpretability analysis and per-city/per-star-rating breakdowns. (2) Practical utility via retrospective simulation - a pricing opportunity analysis for the hotelier view (quantifying cases where a hotel’s listed price deviated from the predicted market movement during high-demand periods, i.e., potential mispricing detected by the system), and a booking savings simulation for the consumer view (savings achieved by following forecast-based timing versus immediate booking). Expected deliverables: the curated Vietnamese hotel price dataset, four trained and compared ML models with full experimental analysis, and a deployable dual-view web application.